



Start with ifcSQL_Tools QGIS plugin

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MSSQL

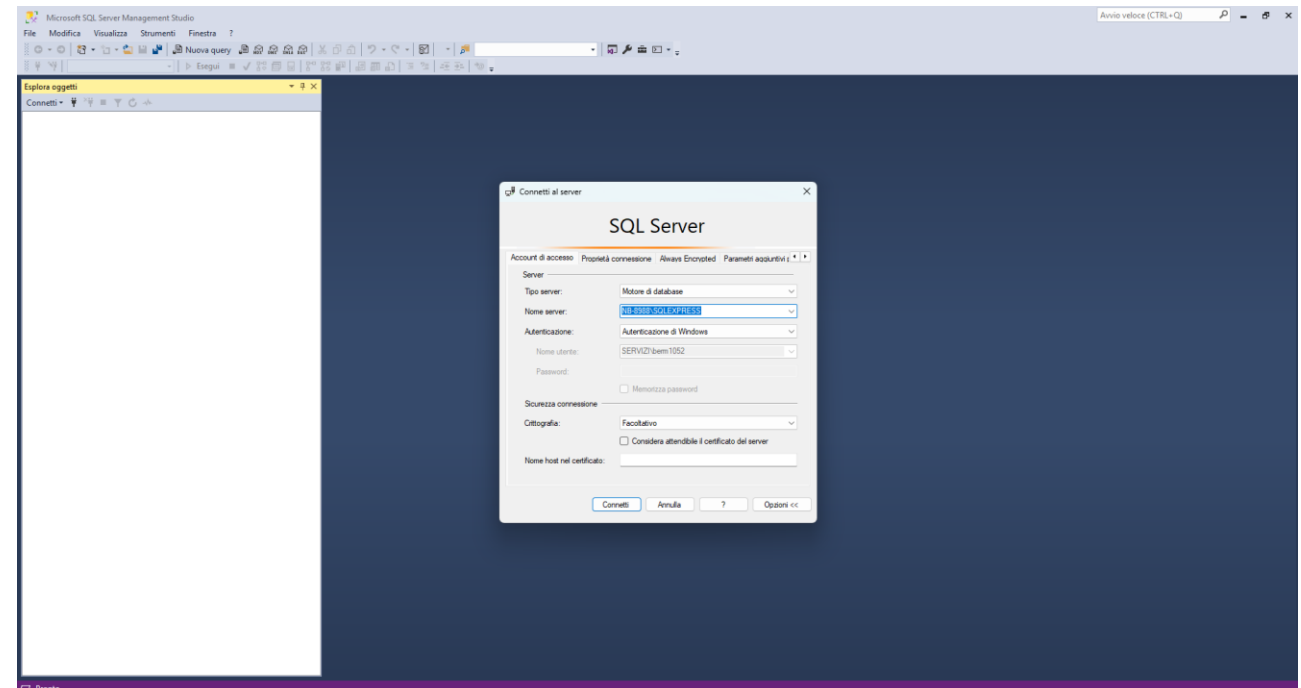


Download and install Microsoft SQL server (MSSQL) and SQL Server Management Studio (SSMS)

Microsoft SQL Server Express 22
(<https://www.microsoft.com/it-it/download/details.aspx?id=104781>)

SQL Server Management Studio 20
(<https://learn.microsoft.com/it-it/ssms/release-notes-20>)

It is important to download and install these versions.





PostgreSQL



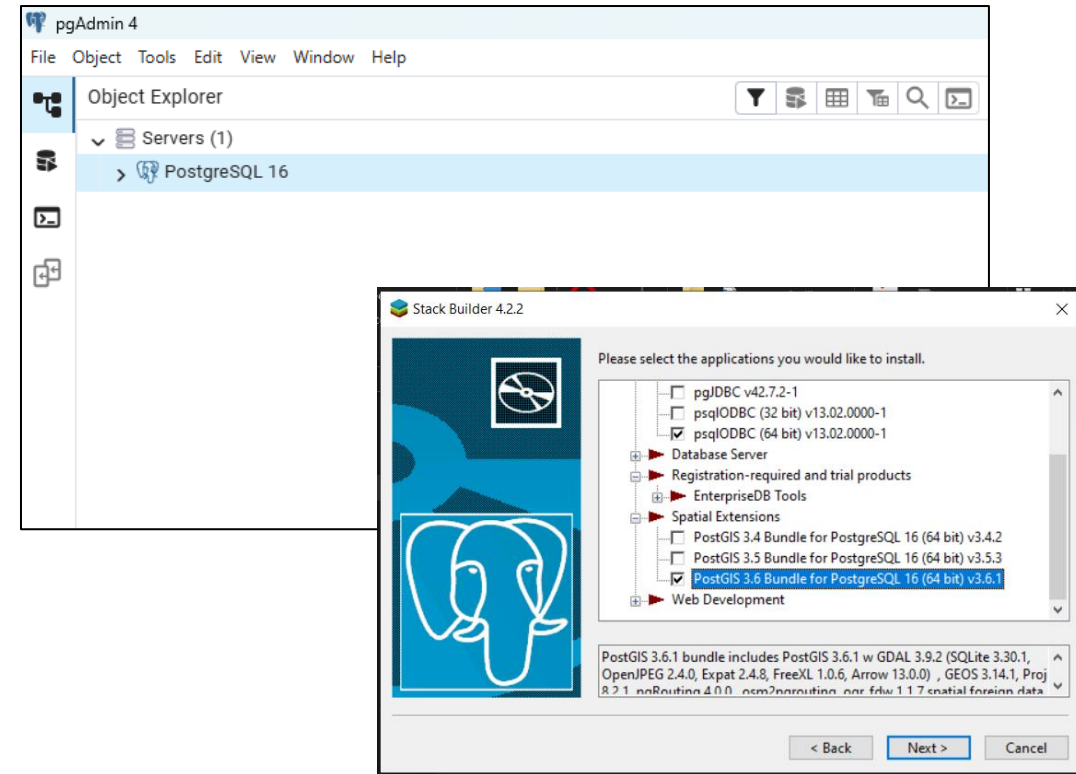
Download and install PostgreSQL and pgAdmin

You can download and install everything together:

<https://www.enterprisedb.com/downloads/postgres-postgresql-downloads>

It is important to download and install the version: **PostgreSQL 16**.

Remember to activate PostGIS during the installation.

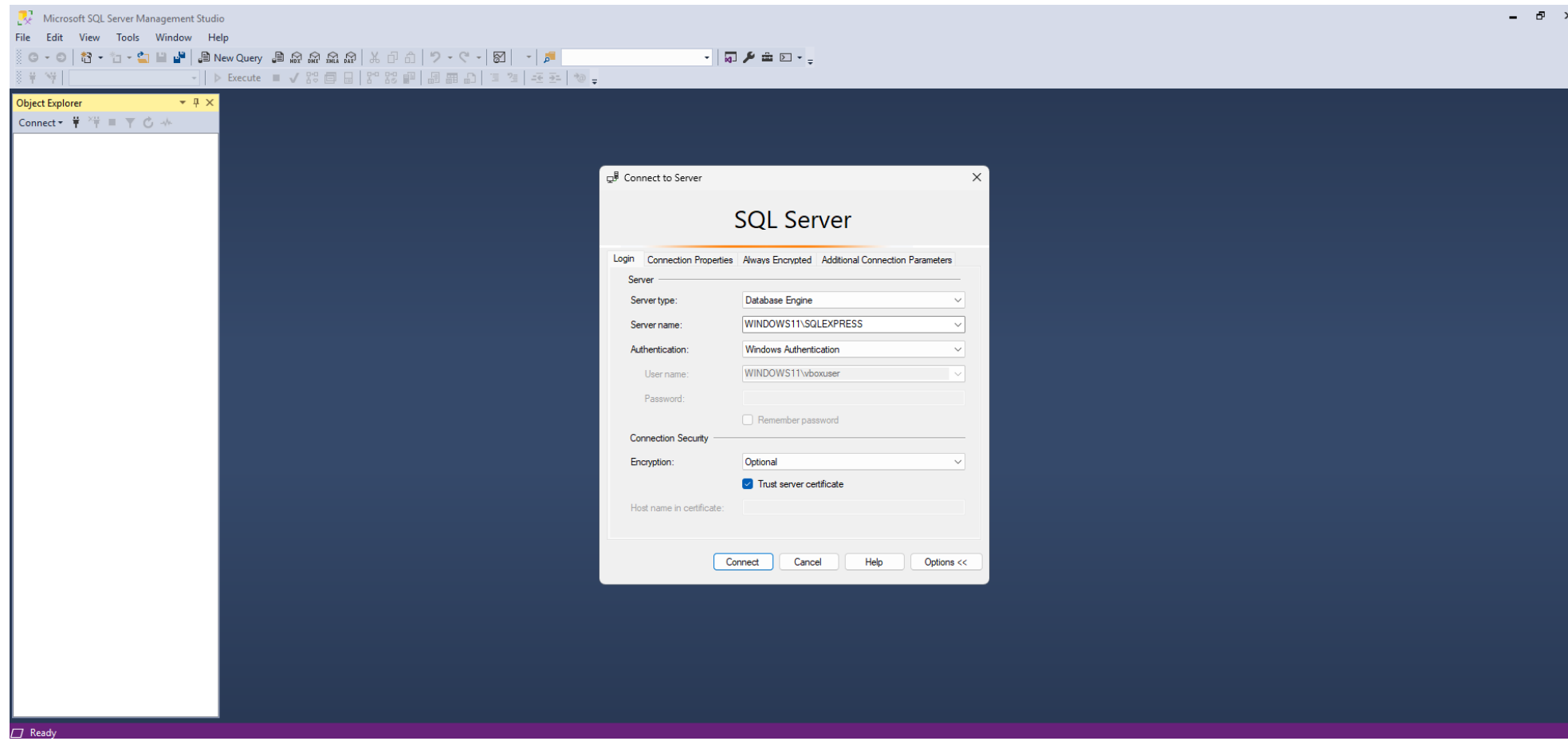




Connect to MSSQL



Connect to MSSQL using “Windows authentication”.





Click the “1.ifcSQL_Installation_MSSQL.bat” file to install ifcSQL on MSSQL

Press a key until the process is finish.

The process may take some minutes.

```
ifcSQL Database Installation

=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: DESKTOP-D9BTGCS\SQLEXPRESS01_
```



ifcSQL_Installation_MSSQL



Click on the bat file.

1

```
ifcSQL Database Installation

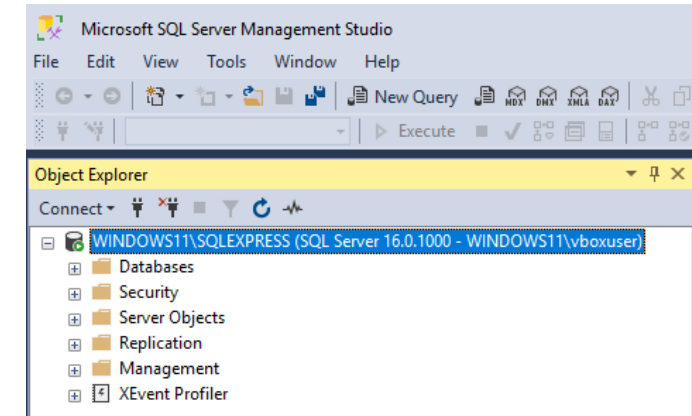
=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name:
```

Retrieve the server name from MSSQL.

2



Paste the server name.

3

```
ifcSQL Database Installation

=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: Windows11\SQLEXPRESS
```

Press a key until the process is finished.

4

```
ifcSQL Database Installation

=====
ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: Windows11\SQLEXPRESS

[INFO] Testing connection to 'Windows11\SQLEXPRESS'...
[SUCCESS] Connected successfully to the SQL Server!

=====
Server set to: Windows11\SQLEXPRESS
Starting the procedure...

Note: The procedure will pause several times to show progress.
Press "Enter" whenever prompted to continue.

=====
--- PHASE 1: Database Creation ---
Press any key to continue . . .
```



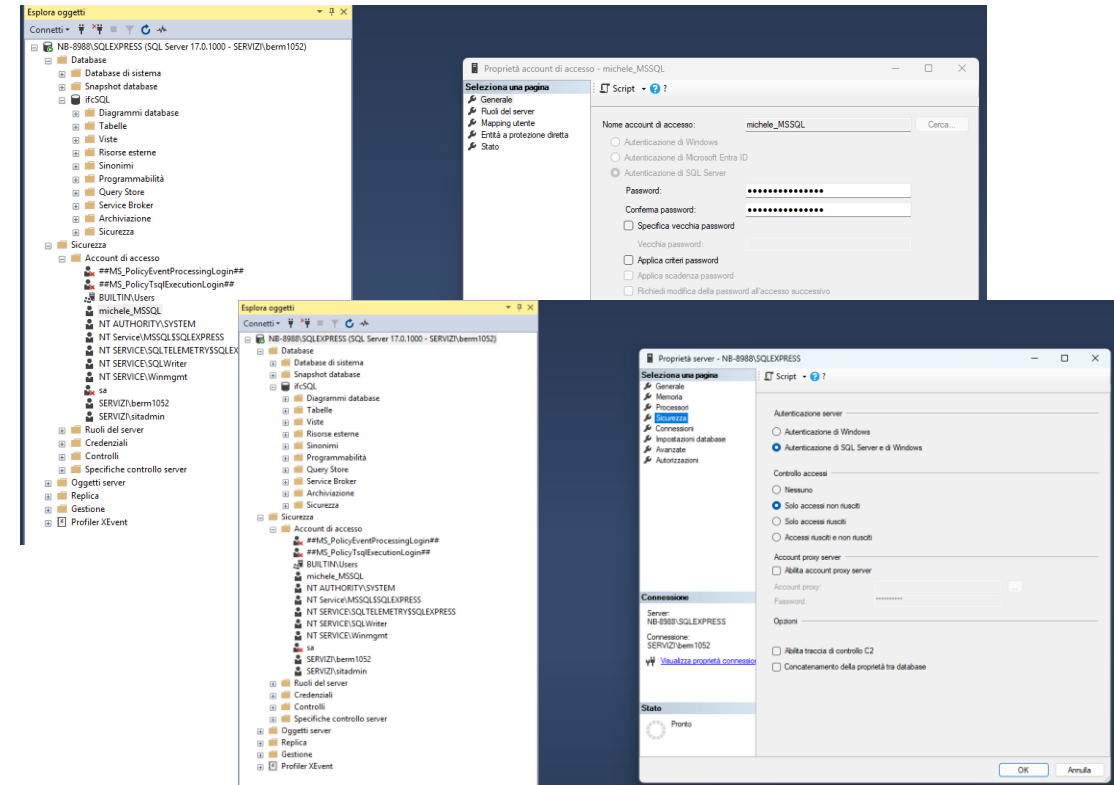

Create a USER in MSSQL



Follow the “**2.Create a user in MSSQL**” guideline.

Summary of the step:

- Create an MSSQL user with SQL server authentication and password.
- Give the user administrator privileges.
- Set up access with mixed authentication (SQL server + Windows).
- Verify that the user can log in to MSSQL.





Click the “3.ifcSQL_Mirror_Installation_PostgreSQL.bat” file to set up the PostgreSQL database

REMEBER: run as administrator the bat file.

Follow the instruction in the bat.

Check at the end to make sure everything is working; you may have entered the wrong credentials.

```
Administrator: ifcSQL Database Setup and TDS FDW Installation
=====
PostgreSQL 16 Database Setup & FDW Installation
=====
[INFO] PostgreSQL base path found at: C:\Program Files\PostgreSQL\16
[INFO] Deploying TDS FDW files to PostgreSQL directories...
Copying extension files (.control, .sql)...
Updating module_pathname in tds_fdw.control...
Stopping PostgreSQL service to unlock DLL files...
Copying DLL binaries to bin folder...
Copying tds_fdw.dll to lib folder...
Restarting PostgreSQL service...
[SUCCESS] FDW files deployed successfully!

--- Database Creation ---
Enter PostgreSQL username (e.g., postgres):
```

```
--- Connection Test ---
Attempting to read the first 5 rows from 'ifcinstance.entity'...
ERROR: DB-Library error: DB #: 20009, DB Msg: Unable to connect: Adaptive Server is unavailable or does not exist (Windows11\SQLEXPRES), OS #: 0, OS Msg: No error, Level: 9

[ERROR] Could not read from MSSQL. Check if the MSSQL server is reachable and credentials are correct (server name; port; username; password).

Do you want to re-enter the MSSQL credentials and try again? (Y/N): y
```



Enter PostgreSQL username and password.

1

```
Amministratore: ifcSQL Database Setup and TDS FDW Installation

=====
PostgreSQL 16 Database Setup & FDW Installation
=====

[INFO] PostgreSQL base path found at: C:\Program Files\PostgreSQL\16

[INFO] Deploying TDS FDW files to PostgreSQL directories...
Copying extension files (.control, .sql)...
Updating module_pathname in tds_fdw.control...
Stopping PostgreSQL service to unlock DLL files...
Copying DLL binaries to bin folder...
Copying tds_fdw.dll to lib folder...
Restarting PostgreSQL service...
[SUCCESS] FDW files deployed successfully!

--- PostgreSQL Authentication ---
Enter PostgreSQL username (e.g., postgres): postgres
Enter password for user postgres: 
```

N.B. When you install PostgreSQL, it asks you to set a password, that is the one you need to enter here. Instead, the username that is automatically created is usually “postgres”.

Enter MSSQL server name, port, username and password.

2

```
Amministratore: ifcSQL Database Setup and TDS FDW Installation

--- MSSQL Foreign Data Wrapper Setup ---
Enter MSSQL Server Name without SQLEXPRESS (e.g., NB-8988): DESKTOP-D9BTGCS
Enter MSSQL Port [Press Enter for 1433]: 1433
Enter MSSQL Username (e.g., mssql_user): mssql_user
Enter MSSQL Password: 
```

Check if the connection works.

3

```
Amministratore: ifcSQL Database Setup and TDS FDW Installation

--- Connection Test ---
Attempting to read the first 5 rows from 'ifcinstance.entity'...
NOTIFICA: tds_fdw: Query executed correctly
NOTIFICA: tds_fdw: Getting results
GlobalEntityInstanceId | EntityTypeId
-----+-----
1 | 400991

(1 riga)

[SUCCESS] Everything is working perfectly! Connection to MSSQL established.
Premere un tasto per continuare . . .
```



Ready to start



You are ready to start using the plugin!
(find out how to use it on GitHub)



Bonus



Would you like to *optionally* set up a second or new database system (MSSQL + PostgreSQL)?



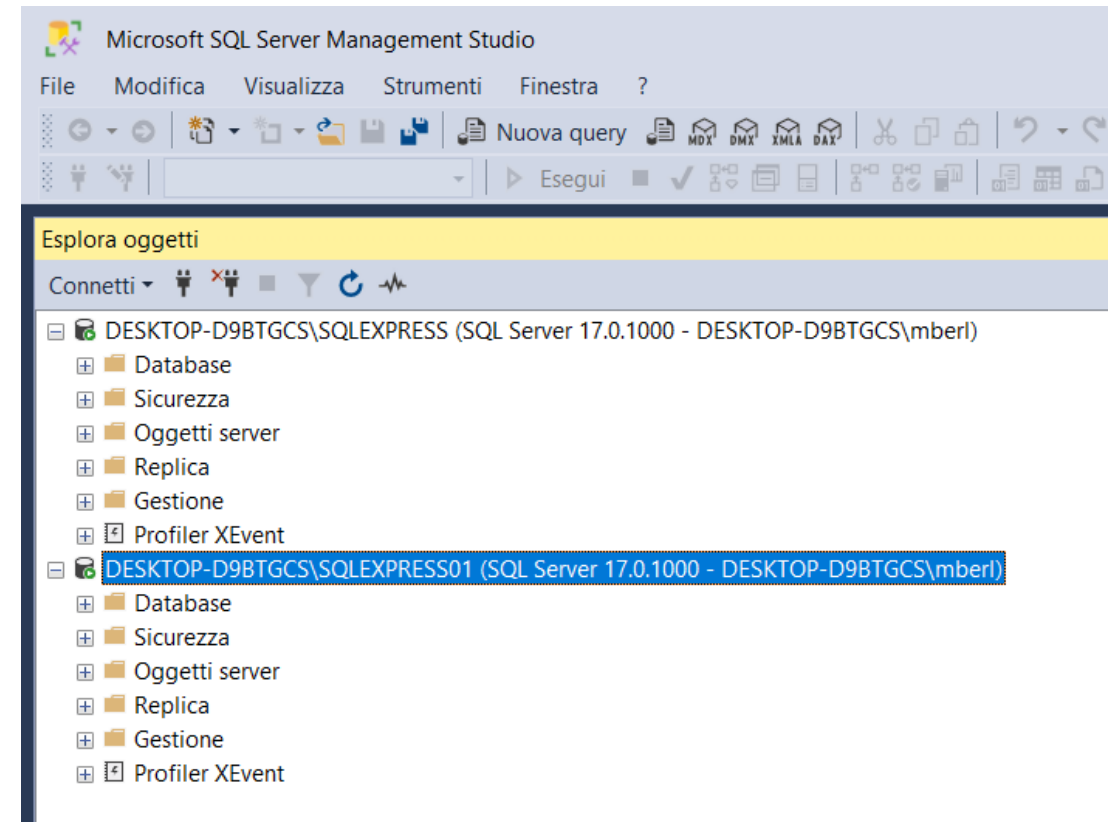
MSSQL



Install **again** Microsoft SQL server (MSSQL)

Microsoft SQL Server Express 22
(<https://www.microsoft.com/it-it/download/details.aspx?id=104781>)

It is important to set up a new server for the second database.





Click the “**1.ifcSQL_Installation_MSSQL.bat**” file to install ifcSQL on MSSQL

This time, use the SQL Server Name of the **second server**.

```
ifcSQL Database Installation

=====
  ifcSQL Database Configuration and Installation
=====

Format example: PC-NAME\SQLEXPRESS

Enter your SQL Server name: DESKTOP-D9BTGCS\SQLEXPRESS01_
```



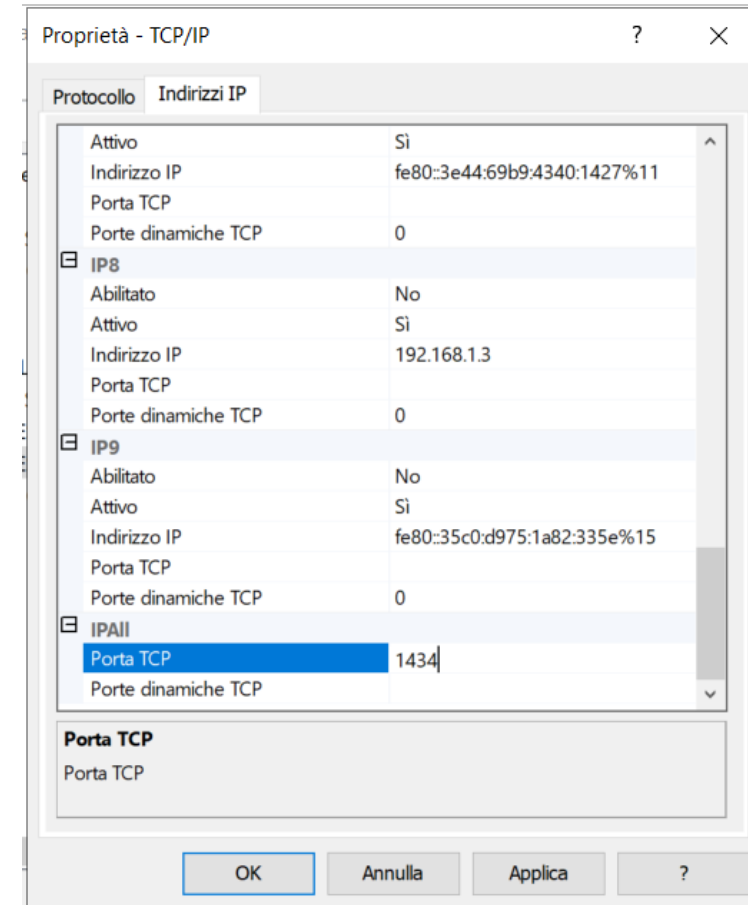
Create a USER in MSSQL



Follow the “**2.Create a user in MSSQL**” guideline.

For this second server, be sure to set a **different TCP port!**

For example: 1434.





Click the “3.ifcSQL_Mirror_Installation_PostgreSQL.bat” file to set up the PostgreSQL database

REMEBER: run as administrator the bat file.

Answer “Y” to the question “Do you want to create a new numbered database?”

```
Administrator: ifcSQL Database Setup and TDS FDW Installation
=====
PostgreSQL 16 Database Setup & FDW Installation
=====

[INFO] PostgreSQL base path found at: C:\Program Files\PostgreSQL\16

[INFO] Deploying TDS FDW files to PostgreSQL directories...
Copying extension files (.control, .sql)...
Updating module_pathname in tds_fdw.control...
Stopping PostgreSQL service to unlock DLL files...
Copying DLL binaries to bin folder...
Copying tds_fdw.dll to lib folder...
Restarting PostgreSQL service...
[SUCCESS] FDW files deployed successfully!

--- Database Creation ---
Enter PostgreSQL username (e.g., postgres): postgres
Enter password for user postgres: 5432

[INFO] The database 'ifcSQL' already exists.
Do you want to create a new numbered database? (Y/N): _
```




The end



You can repeat the procedure as many times as you want.